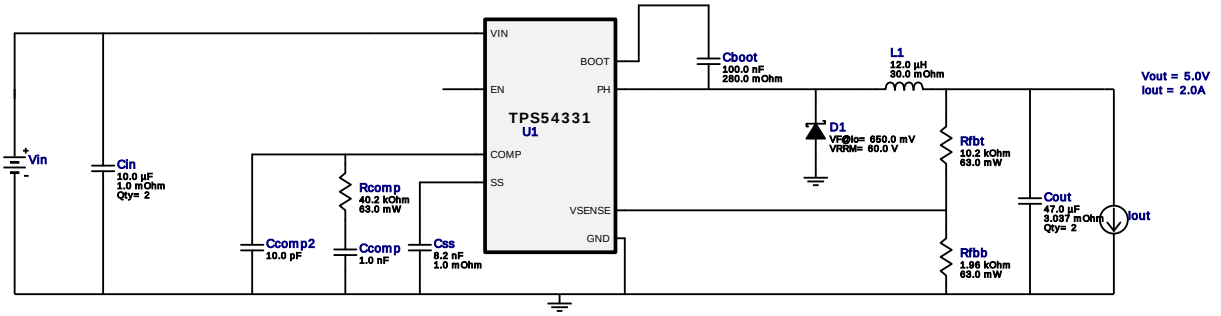
WEBENCH ® Design Report

VinMin = 14.0V
VinMax = 26.0V
Vout = 5.0V
Iout = 2.0A

Device = TPS54331DDAR
Topology = Buck
Created = 2026-02-09 07:38:30.798
BOM Cost = $2.21
BOM Count = 14
Total Pd = 1.45W

Design : 8 TPS54331DDAR
TPS54331DDAR 14V-26V to 5.00V @ 2A



Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccomp	MuRata	GRM1555C1H102JA01J Series= C0G/NP0	Cap= 1.0 nF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccomp2	Yageo	CC0805JRNPO9BN100 Series= C0G/NP0	Cap= 10.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cin	TDK	C3225X7R1H106M250AC Series= X7R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 5.0 A	2	$0.27	1210 15 mm²
Cout	MuRata	GRM32ER61C476KE15L Series= X5R	Cap= 47.0 uF ESR= 3.037 mOhm VDC= 16.0 V IRMS= 4.59346 A	2	$0.17	1210_280 15 mm²
Css	MuRata	GRM033R71A822KA01D Series= X7R	Cap= 8.2 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
D1	Fairchild Semiconductor	FSV360FP	VF@Io= 650.0 mV VRRM= 60.0 V	1	$0.13	SOD-123HE 13 mm²
L1	Bourns	SDR1307-120ML	L= 12.0 µH 30.0 mOhm	1	$0.51	SDR1307 226 mm²
Rcomp	Vishay-Dale	CRCW040240K2FKED Series= CRCW..e3	Res= 40.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbb	Vishay-Dale	CRCW04021K96FKED Series= CRCW..e3	Res= 1.96 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbt	Vishay-Dale	CRCW040210K2FKED Series= CRCW..e3	Res= 10.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²

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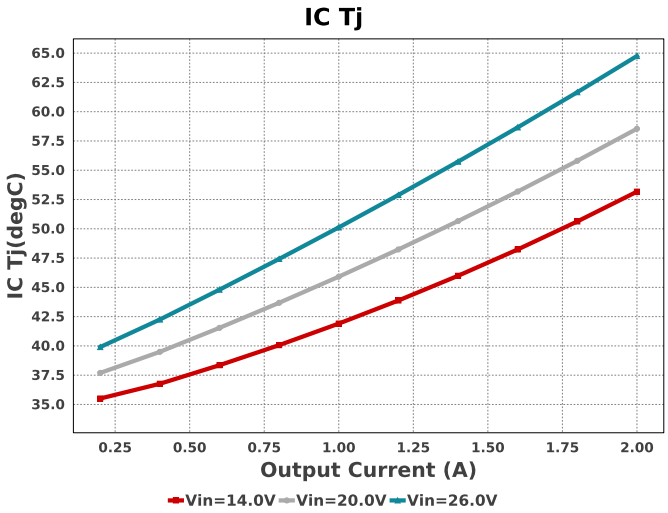
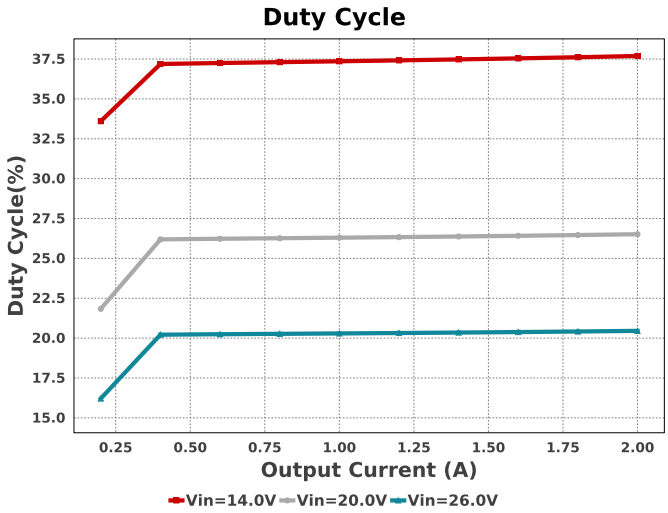
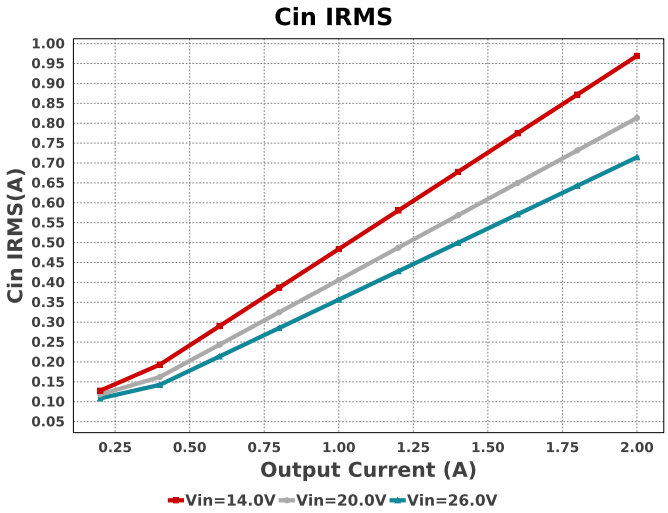
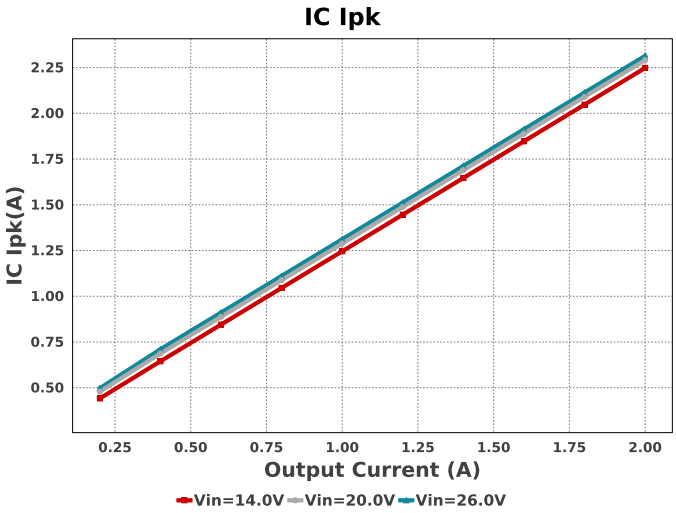
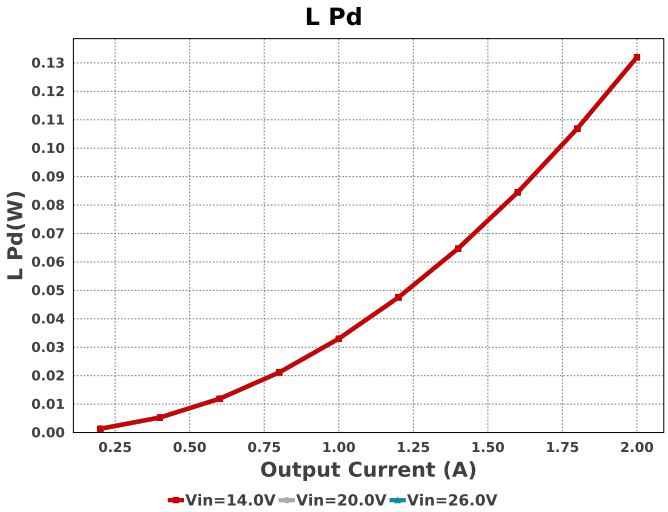
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
U1	Texas Instruments	TPS54331DDAR	Switcher	1	$0.62	DDA0008H 55 mm²

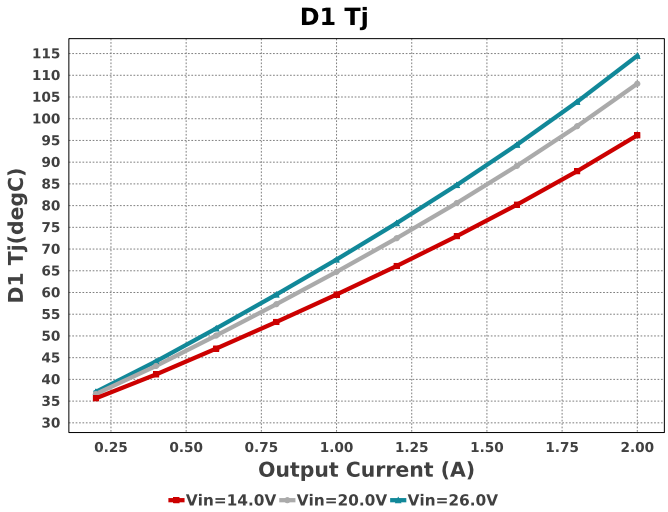












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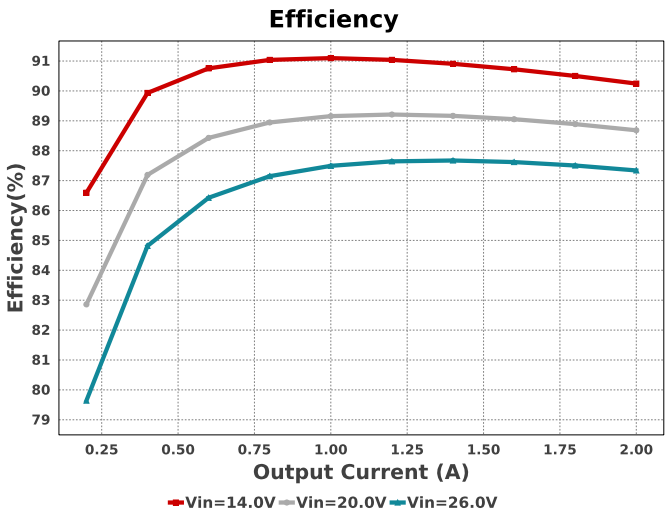

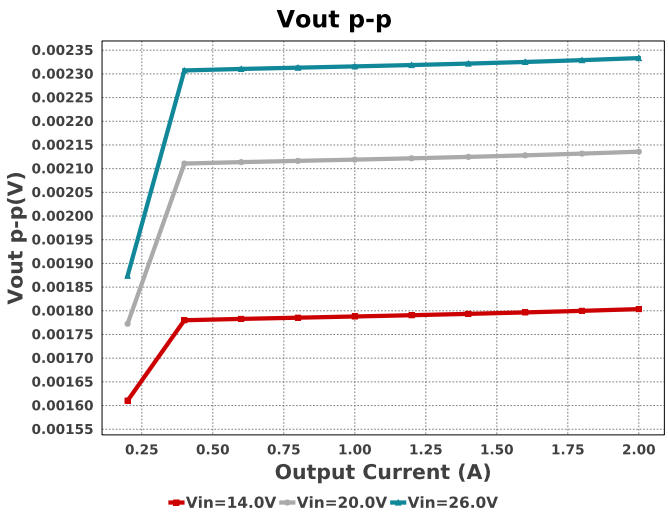

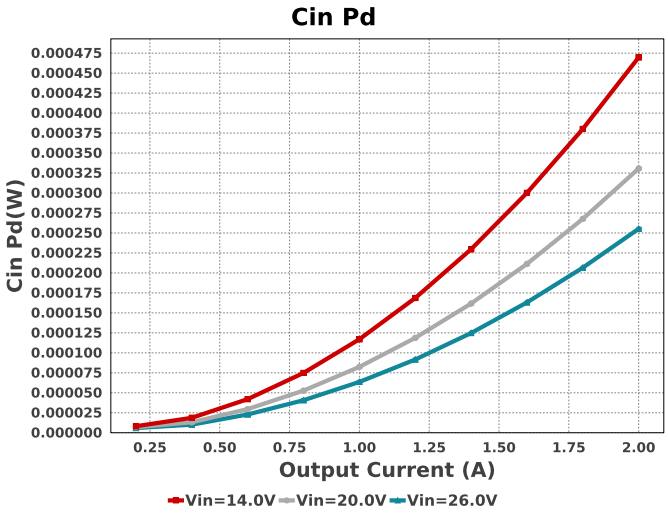

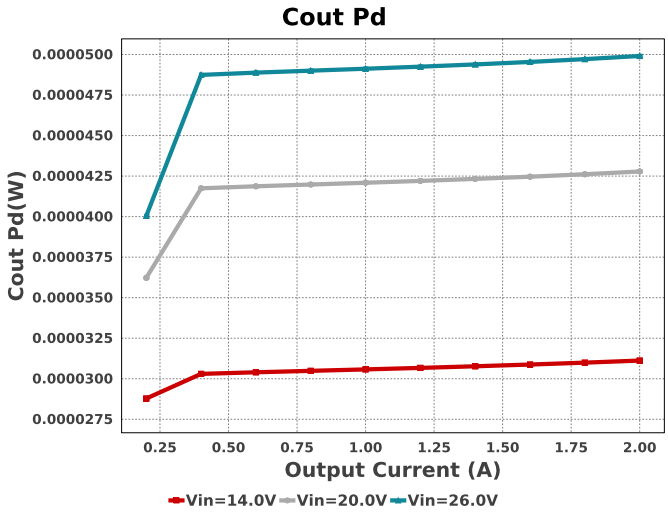

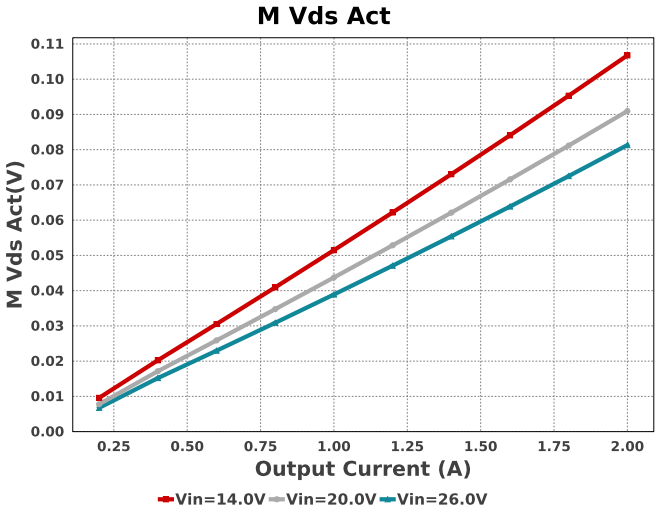

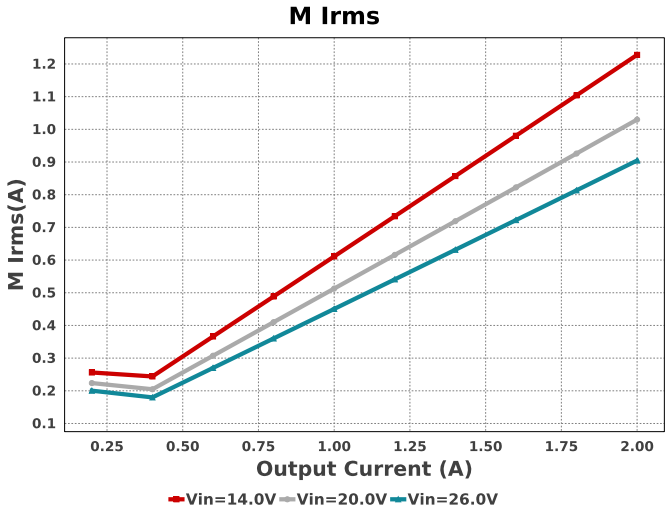

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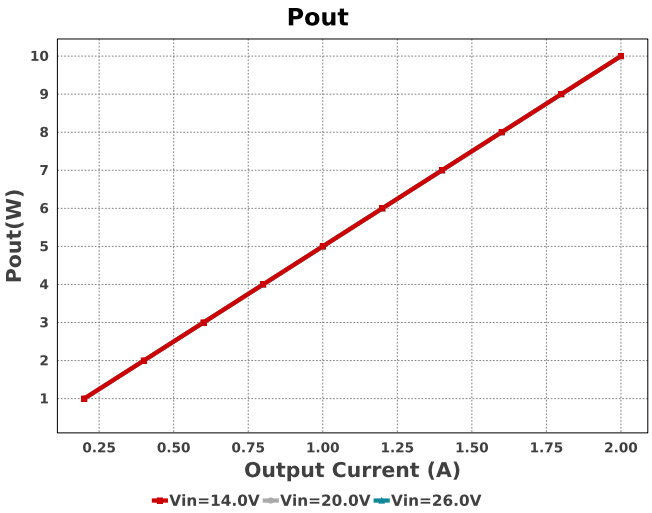
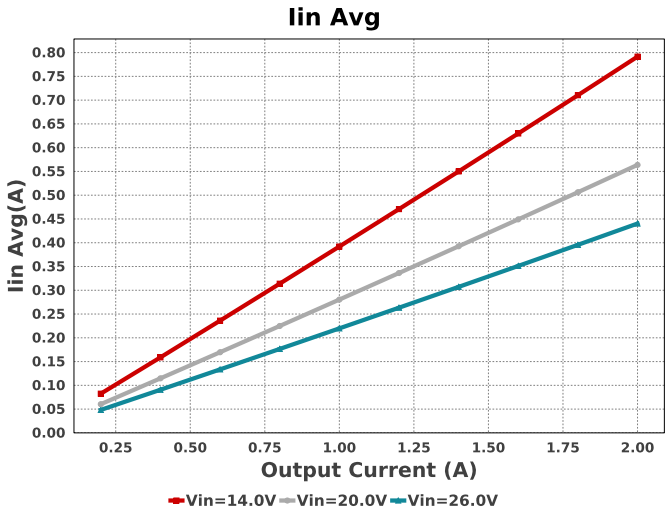
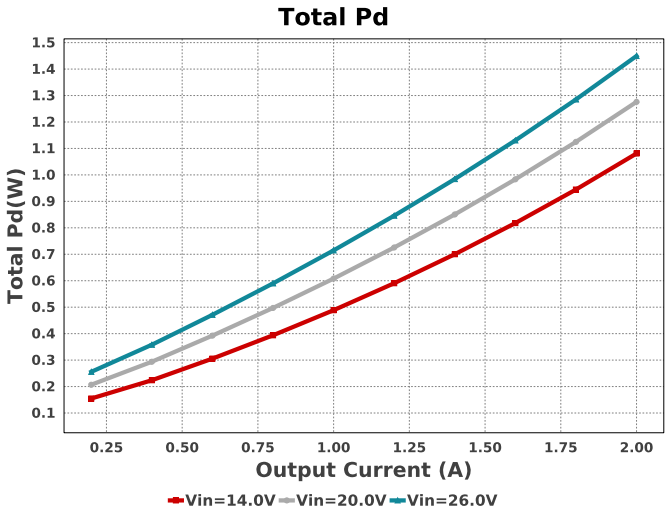
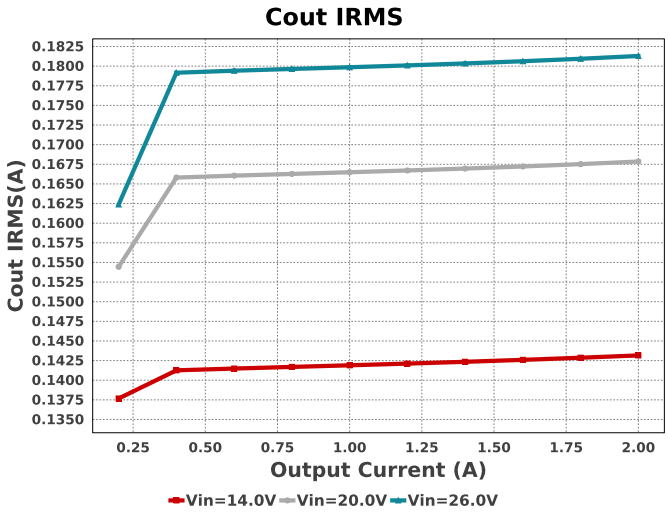

Pout
Pout(W)
Output Current (A)
Vin=14.0V Vin=20.0V Vin=26.0V


Iin Avg
Iin Avg(A)
Output Current (A)
Vin=14.0V Vin=20.0V Vin=26.0V


Total Pd
Total Pd(W)
Output Current (A)
Vin=14.0V Vin=20.0V Vin=26.0V


Cout IRMS
Cout IRMS(A)
Output Current (A)
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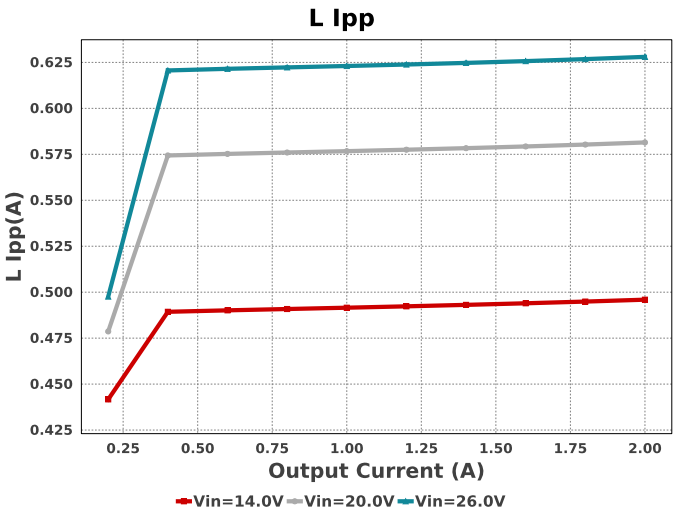
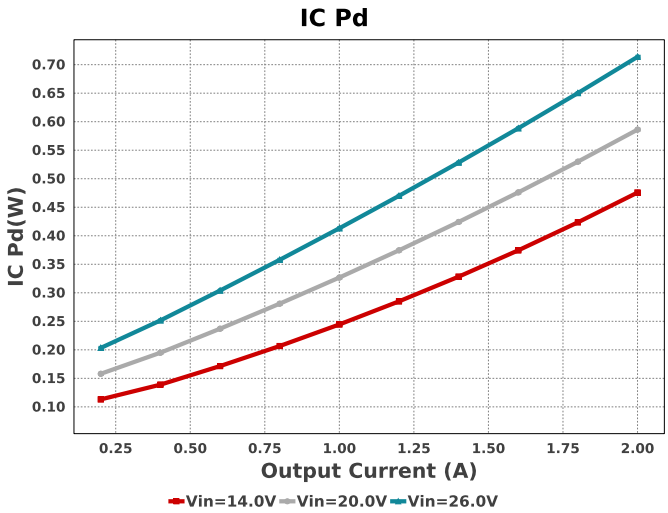
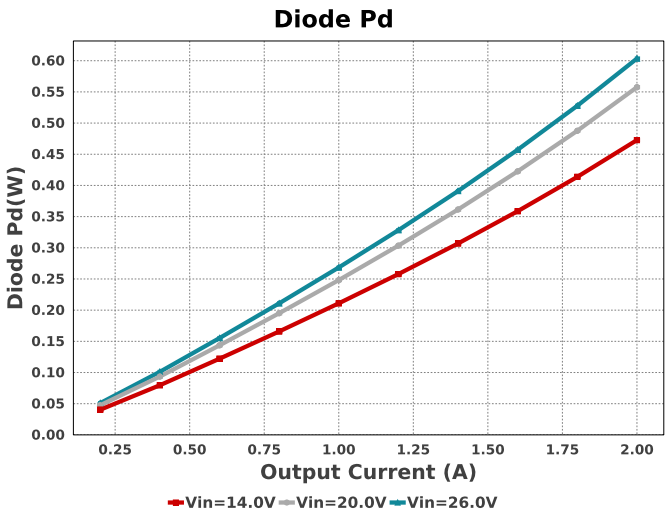
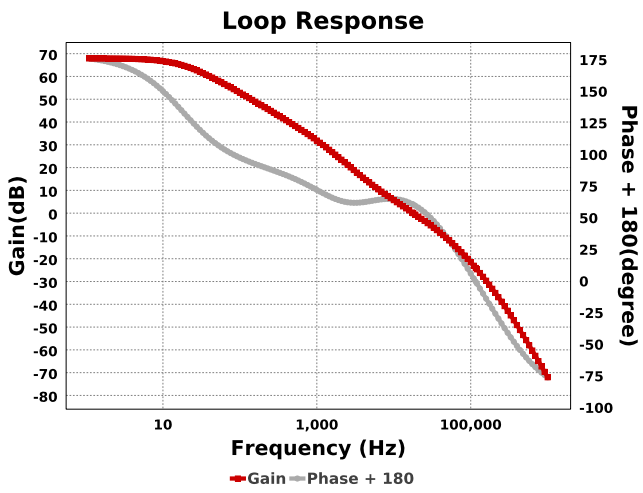

L Ipp
L Ipp(A)
Output Current (A)
Vin=14.0V Vin=20.0V Vin=26.0V


IC Pd
IC Pd(W)
Output Current (A)
Vin=14.0V Vin=20.0V Vin=26.0V





Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	714.407 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	255.19 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	181.292 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	49.908 µW	Capacitor	Output capacitor power dissipation
5.	D1 Tj	114.49 degC	Diode	D1 junction temperature
6.	Diode Pd	603.5 mW	Diode	Diode power dissipation
7.	IC Ipk	2.314 A	IC	Peak switch current in IC
8.	IC Pd	713.67 mW	IC	IC power dissipation
9.	IC Tj	64.756 degC	IC	IC junction temperature
10.	ICThetaJA	48.7 degC/W	IC	IC junction-to-ambient thermal resistance
11.	Iin Avg	440.36 mA	IC	Average input current
12.	L Ipp	628.01 mA	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	132.0 mW	Inductor	Inductor power dissipation
14.	M Irms	904.551 mA	Mosfet	MOSFET RMS ripple current
15.	M Vds Act	81.308 mV	Mosfet	Voltage drop across the MosFET
16.	Cin Pd	255.19 µW	Power	Input capacitor power dissipation
17.	Cout Pd	49.908 µW	Power	Output capacitor power dissipation
18.	Diode Pd	603.5 mW	Power	Diode power dissipation
19.	IC Pd	713.67 mW	Power	IC power dissipation
20.	L Pd	132.0 mW	Power	Inductor power dissipation
21.	Total Pd	1.449 W	Power	Total Power Dissipation
22.	BOM Count	14	System Information	Total Design BOM count
23.	Cross Freq	17.902 kHz	System Information	Bode plot crossover frequency
24.	Duty Cycle	20.455 %	System Information	Duty cycle
25.	Efficiency	87.34 %	System Information	Steady state efficiency
26.	FootPrint	381.0 mm^2	System Information	Total Foot Print Area of BOM components
27.	Frequency	570.0 kHz	System Information	Switching frequency
28.	Gain Marg	-24.057 dB	System Information	Bode Plot Gain Margin
29.	Iout	2.0 A	System Information	Iout operating point
30.	Low Freq Gain	67.904 dB	System Information	Gain at 1Hz
31.	Mode	CCM	System Information	Conduction Mode
32.	Phase Marg	60.974 deg	System Information	Bode Plot Phase Margin
33.	Pout	10.0 W	System Information	Total output power
34.	Total BOM	$2.21	System Information	Total BOM Cost
35.	Vin	26.0 V	System Information	Vin operating point
36.	Vout	5.0 V	System Information	Operational Output Voltage

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#	Name	Value	Category	Description
37.	Vout Actual	4.963 V	System Information	Vout Actual calculated based on selected voltage divider resistors
38.	Vout Tolerance	5.254 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
39.	Vout p-p	2.333 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	2.0	Maximum Output Current
VinMax	26.0	Maximum input voltage
VinMin	14.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	TPS54331	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

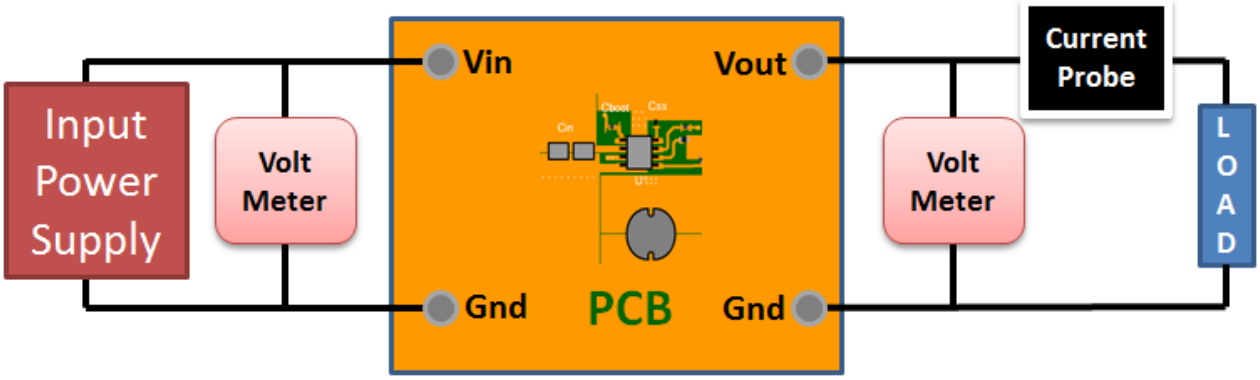
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 29DE6FE5A8ACC9ACBA48DC298604997C[v1]
2. **TPS54331** Product Folder : http://www.ti.com/product/TPS54331 : contains the data sheet and other resources.

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